2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)

Exploring Submerged Cultural Heritage in VR through Hand-based Multimodal Haptic Interaction

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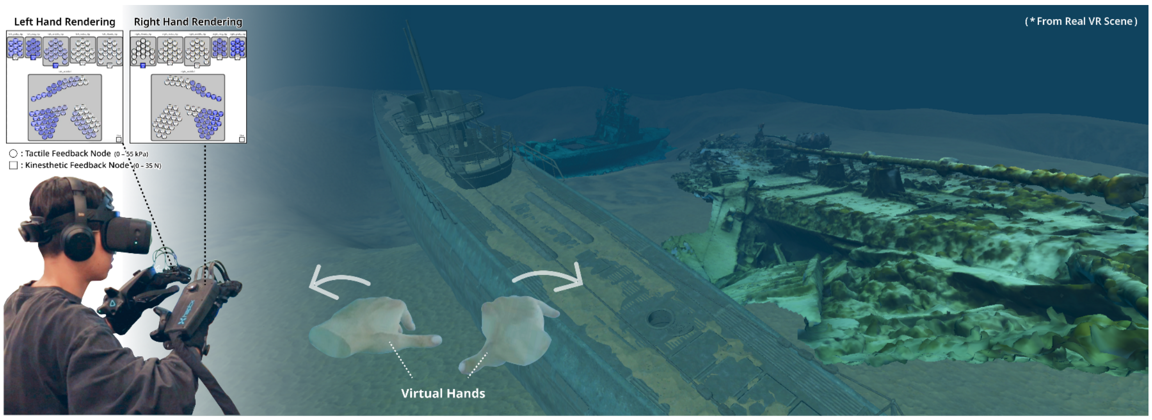

Figure 1: A user wearing pneumatic haptic gloves explores a submerged shipwreck cultural heritage site in a virtual environment [1]. The system provides multimodal haptic feedback incorporating drag and added mass forces, dynamically modulated by hand movement and individual finger posture. This haptic interaction plays a critical role in shaping immersive underwater experiences.

Abstract

Exploring submerged cultural heritage, such as shipwrecks and underwater archaeological sites, involves multisensory experiences with aquatic environments. While virtual reality (VR) provides a non-invasive way to experience submerged heritage, existing VR-based representations largely prioritize visual involvement, offering limited support for material and dynamic aspects of underwater interaction. In this work, we present a hand-based multimodal haptic interaction system for immersive exploration of submerged cultural heritage in VR. By focusing on human-perceivable forces, our approach balances perceptual fidelity and computational efficiency through selective rendering of drag and added mass forces arising from fluid-hand interaction. Using pneumatic-based haptic gloves, tactile and kinesthetic feedback are dynamically adapted to the user's hand movement and posture. This work advances experience-oriented digital heritage systems by supporting embodied interaction with submerged archaeological environments beyond visual depiction alone.

Index Terms: Underwater Cultural Heritage, Multimodal Haptics, Fluid-Haptic Interaction, Virtual Reality.

1 Introduction

Submerged cultural heritage sites, such as shipwrecks, underwater architectural and settlements remain, constitute a critical source for interpreting past human cultures and their interaction with aquatic environments [2, 3]. However, direct access to such sites is often constrained by environmental, safety, and conservation challenges [4].

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To address these challenges, immersive extended reality (XR) technologies have been increasingly adopted to digitally reconstruct and present underwater cultural heritage, enabling non-invasive exploration [5, 6]. Such approaches support a range of use cases, including archaeology education [7], museum experiences [8], and diver training [9, 10]. Prior work has demonstrated that virtual reality (VR) [11, 12] and augmented reality (AR) [13, 14] systems can enhance the accessibility and interpretability of submerged sites by providing visual context and spatial understanding. These systems allow users to engage with digital reconstructions through interaction and mediated sensory feedback, thereby supporting richer experiential understanding and cognitive engagement.

Despite these advances, most XR-based heritage experiences focus primarily on visual realism [15, 16], offering limited support for conveying the material and dynamic nature of underwater environments. Because underwater movement is inherently shaped by the fluid resistance, visual feedback alone is often insufficient to evoke a sense of presence and immersion. While various haptic interfaces have been proposed to enrich embodied interaction in XR [17, 18], the design space for haptic interaction with virtual submerged environments remains relatively underexplored [19]. This motivates us to outline an approach that explores virtual fluid interaction through human-perceivable force cues.

In this paper, we argue that hand-based multimodal haptic interaction plays a critical role in shaping underwater cultural heritage experiences. Rather than relying on computationally expensive fluid simulations, we adopt a user-centered rendering approach that prioritizes perceptible fluid forces [20, 21]. When users move their hands underwater, the system delivers drag and added mass forces in response to hand movement and finger posture [22]. By integrating this method into a hand-based haptic system, our work balances physical plausibility with perceptual effectiveness, supporting immersive exploration of submerged heritage environments.

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DOI 10.1109/VRW70859.2026.00013

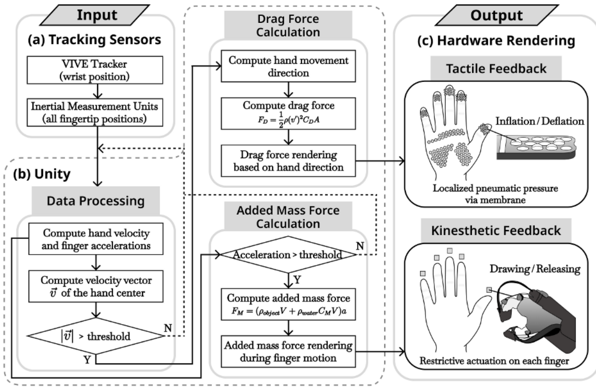

Figure 2: The whole framework of the multimodal haptic rendering system. (a) Acquiring hand positions from tracking sensors. (b) Computing fluid forces based on the detected hand and finger motions. (c) Rendering multimodal haptic feedback through each tactile and kinesthetic actuator.

2 Haptic Rendering System

In our system, the physical properties of a virtual underwater environment are delivered through haptic feedback in response to the user's motions. The following describes the assumed fluid model, the mapping between fluid properties and haptic actuators, and the overall system workflow.

2.1 Virtual Underwater Configuration

Accurately simulating fluid dynamics in real-time remains computationally expensive [19, 23], particularly when combined with interactive VR systems. From this perspective, we focus on two primary perceivable cues. When humans move their hands underwater, drag and added mass forces are generated in response to motion. The drag force acts normal to the hand surface, while the added mass force arises from finger acceleration [24]. We consider a virtual underwater environment composed of stationary fluid, where only the user's hand is in motion. When the hand stops moving, the surrounding fluid is assumed to remain stationary [25, 26].

2.2 Multimodal Actuator Assignment

We present a multimodal haptic rendering system that reflects key aspects of underwater materials by adapting to the user's hand movement and posture. The system is implemented using wearable pneumatic-based haptic gloves (DK2, HaptX Inc.[1]), which provide localized tactile feedback via controllable air pressure and kinesthetic feedback through wire-driven resistive actuation.

Because drag forces act normal to the surface of the hand, tactile feedback is used to convey the resistive properties of the fluid. Since sensations related to fluid inertia and apparent weight are predominantly perceived through force-based cues, added mass forces are therefore rendered using kinesthetic feedback [27].

2.3 System Architecture

As shown in Fig. 2, our system architecture integrates three primary components: (a) tracking sensor data embedded in the haptic gloves, (b) computing hand velocity, acceleration, and fluid-induced forces based on the user's motion within the Unity Engine, and (c) mapping the computed forces to individual actuators via haptic gloves for hardware rendering.

The system continuously tracks the user's hand movement and finger posture at each frame. Instead of detecting collision events with fluid particles, haptic feedback is generated in direct response to the user's movement within the virtual fluid environment.

In the drag force formulation, ρ denotes the water density (approximately 998.6 kg/m^3 at 18°C), and v' represents the relative velocity between the hand and the fluid flow (here is the pure hand velocity). The drag coefficient C_D depends on the object's shape, and A denotes the projected cross-sectional area normal to the flow. Based on the specifications of the tactile actuators, we set C_D to 0.42, and A to 6.1 mm for the palm and 4.5 mm for the fingers, corresponding to 9.30×10^{-6} m^2 and 5.06×10^{-6} m^2, respectively.

In the added mass force formulation, ρ_{object} and ρ_{water} denote the densities of the object (approximately 1,050 kg/m^3 for a finger in this case) and water, individually. The added mass coefficient C_M depends on the object's shape, and we set it to 1 following simplified cylinder-finger shaper [29]. V represents the object's volume, and a its acceleration. Based on the distal phalanges, the combined mass $(m + m_{added})$ was assigned per finger according to anatomical measurements. The resulting values are on the order of $10^{-6}kg$ and reflect differences in finger size [30].

[1] https://haptx.com/

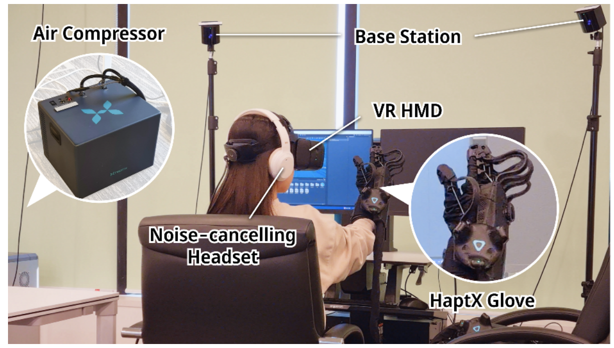

Figure 3: Hardware setup for VR and haptic systems. The user wears HaptX haptic gloves [28], and the air compressor for providing pneumatic-based feedback is placed on the floor.

3 Sensory Simulation through Perceptible Haptics

As shown in Fig. 3, users wear the haptic gloves to experience the proposed system. To enhance immersion, they also wear a headset to reduce the noise from the compressed air and perceive ambient deep-sea audio cues.

Tactile feedback is spatially modulated across the hand to reflect regions against to the fluid-facing parts, while kinesthetic feedback provides motion-dependent resistance. Although the system employs physical formulations to compute fluid forces, our goal is to render forces that are also perceptually meaningful.

When the system detects the direction of hand movement, tactile feedback is delivered accordingly. For example, when the user pushes the fluid with the palm, drag forces are rendered across all tactile actuators on the palm. In contrast, when the user slices through the fluid with a hand blade, drag forces are sequentially rendered across finger position-based tactile actuators according to their spatial arrangement. When there is finger acceleration, added mass forces are rendered through kinesthetic feedback. In this case, users perceive the light mass of the fluid on their fingers through the exoskeletal structure of the glove.

These findings suggest that a relatively simple virtual underwater environment with perceptual hand-based rendering methods can meaningfully improve sensory simulation in heritage VR, without incurring the computational cost of full physical simulation.

4 Discussion

The proposed multimodal haptic interaction system addresses key methodological challenges in the digital representation of submerged cultural heritage. Without relying on precomputed physical simulations or static datasets, the system emphasizes hand-based haptic interaction, allowing tactile and kinesthetic feedback to adapt in real-time to the user's hand motion and posture. This rendering approach supports flexible exploration of virtual underwater environments, which is particularly relevant for archaeological contexts characterized by diverse artifacts, spatial layouts, and embodied modes of interaction.

From a system perspective, our work highlights the importance of integrating XR and haptics for cultural heritage applications. By combining an XR environment with human perceptible forces and haptic feedback, the suggested framework indicates how multisensory interaction can enrich the experiential understanding of archaeological and heritage sites in ways not achievable through visual feedback alone. While the current system assumes a simplified virtual underwater environment, future extensions could incorporate varying conditions of pressure and fluid density to better reflect underwater material properties and support perceptual fidelity.

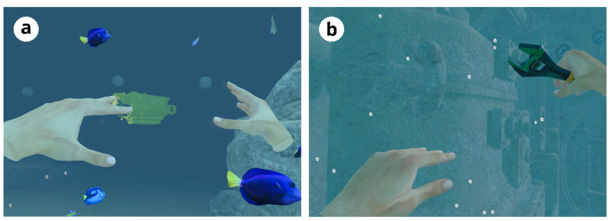

Figure 4: Underwater cultural scenarios including (a) a marine ecosystem and (b) shipwreck restoration training.

In addition to methodological contributions, our system can support a range of future applications in submerged cultural heritage. As illustrated in Fig. 4, it can be applied to educational XR experiences that depict underwater ecosystems, such as navigating seabed environments, where fluid-mediated interaction enhances users' learning and engagement. In addition, the system enables virtual reconstructions of shipwrecks and restoration simulation while overcoming the physical constraints characteristic of underwater activities. These examples demonstrate how multisensory XR systems can extend beyond visual reconstruction to support embodied understanding of submerged heritage contexts.

5 Conclusion

This work presents a hand-based multimodal haptic interaction system for exploring submerged cultural heritage in VR. Motivated by the challenges of accessing and preserving underwater archaeological sites, the proposed approach focuses on hand-perceivable forces to convey the dynamic qualities of underwater interaction. By integrating tactile and kinesthetic feedback into VR environments, the system enables user-centered fluid experiences that foster a stronger sense of immersion during virtual exploration. By aligning multisensory interaction design with the goals of underwater archaeology, this work suggests experience-oriented XR systems that go beyond visual reconstruction alone. Such approaches open new possibilities for engaging with submerged cultural heritage through embodied interaction, supporting richer interpretation and deeper experiential understanding in future digital heritage research.

Acknowledgments

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (IITP-RS-2025-02214780, Generative Haptics and Fine Response Inference for Flexible Tactile Interfaces).

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